

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 221 — Past Questions & Materials
by Renance

Course: MEE 221
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 200

This study pack was compiled and restructured by Renance for MEE 221 students at Federal University of Petroleum Resources, Effurun. It brings together 6 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 6 — MEE 221 ENGINEERING DRAWING I 2023I2024

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING

Second Semester Examination 2023/2024 Session

COURSE CODE AND TITLE: MEE 221 – ENGINEERING DRAWING I

ALL Dimensions are in mm

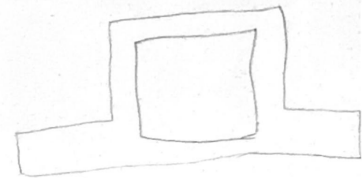
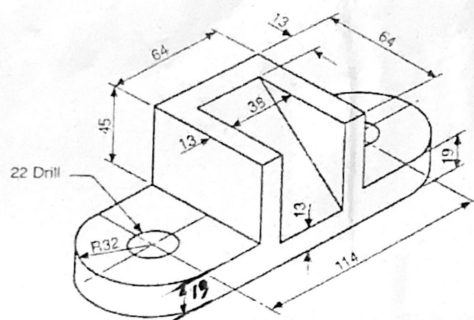
TIME ALLOTTED: 3Hours

INSTRUCTION: ANSWER ANY 3 QUESTIONS

Border line, Title Block - 10marks

QUESTION 1

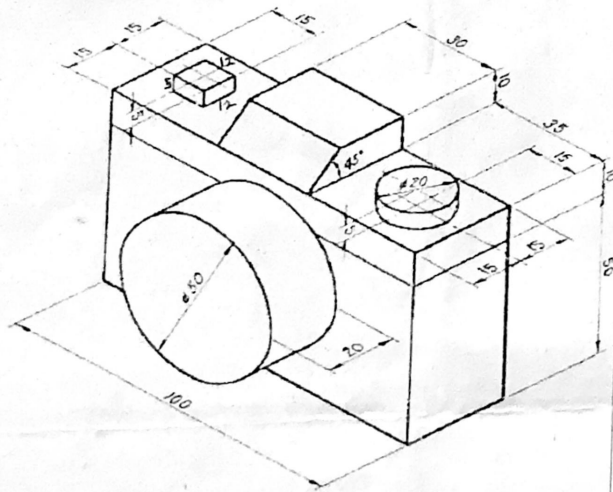
Produce an orthographic projection of the object as shown below, showing the front, plan, and side views in first angle or third angle projection. Indicate the projection used



20marks

QUESTION 2

Make an isometric drawing of the object as shown below



20marks

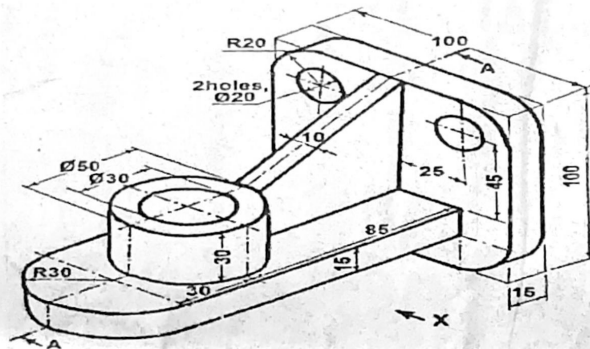
QUESTION 3

Draw the Top and the front view of an M22 bolt and nut having a bolt length of 80 mm and a thread length of 55 mm.

20marks

QUESTION 4

Construct the sectional view of the section A-A as shown in the figure below.



20marks

QUESTION 5

A plate cam rotating clockwise is to give an incline point follower the following motion; $0^\circ - 120^\circ$ lift 36mm uniform velocity, $120^\circ - 180^\circ$ dwell $180^\circ - 360^\circ$ for 36mm with simple harmonic motion. Draw the cam profile if the minimum cam radius is 58mm and the camshaft diameter is 32mm.

20marks

Paper 2 of 6 — MEE 221 Engineering Drawing 1 2024

Level 200 • Second Semester • PastQuestion

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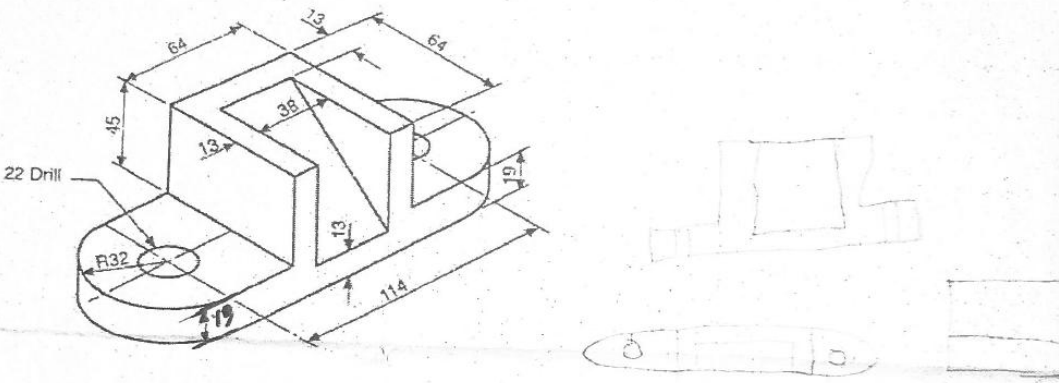
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING
Second Semester Examination 2023/2024 Session
COURSE CODE AND TITLE: MEE 221 – ENGINEERING DRAWING I
TIME ALLOTTED: 3Hours

ALL Dimensions are in mm
INSTRUCTION: ANSWER ANY 3 QUESTIONS

Border line, Title Block - 10marks

✓ **QUESTION 1**

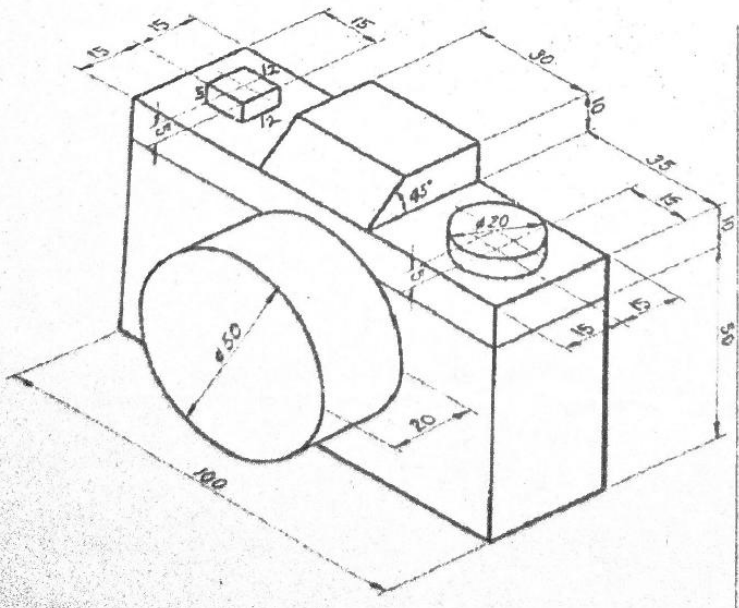
Produce an orthographic projection of the object as shown below, showing the front, plan, and side views in first angle or third angle projection. Indicate the projection used



20marks

QUESTION 2

Make an isometric drawing of the object as shown below



M 22

$$22 \times 1.5 = 33$$

$$22 \times 0.7 = 15.4$$

$$22 \times 0.1 = 2.2$$

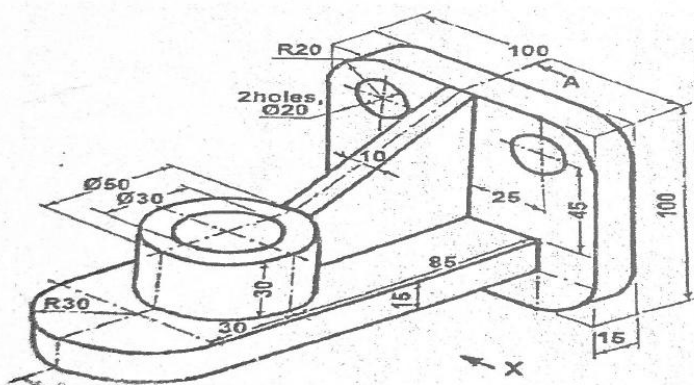
QUESTION 3

Draw the Top and the front view of an M22 bolt and nut having a bolt length of 80 mm and a thread length of 55 mm.

20marks

QUESTION 4

Construct the sectional view of the section A-A as shown in the figure below.



20marks

QUESTION 5

A plate cam rotating clockwise is to give an incline point follower the following motion; $0^\circ - 120^\circ$ lift 36mm uniform velocity, $120^\circ - 180^\circ$ dwell $180^\circ - 360^\circ$ for 36mm with simple harmonic motion. Draw the cam profile if the minimum cam radius is 58mm and the camshaft diameter is 32mm.

20marks

Paper 3 of 6 — MEE 221 -- ENGINEERING DRAWING II (2022/2023)

Level 200 • Second Semester • PastQuestion

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DEPARTMENT OF MECHANICAL ENGINEERING
Second Semester Examination 2022/2023 Session
COURSE CODE AND TITLE: MEE 221 ENGINEERING DRAWING II

Instructions:

- (i). Prepare of border and title block with free hand lettering carefully applied. – 10 marks

Answer Question One and any other two questions. That is a total of Three questions in all.
All dimensions are in mm if the unit is not indicated in the question.

QUESTION 1

30marks

- (a) Produce an assembly drawing of the four components as shown in figure 1.
- (b) Produce a sectional view X-X of the figure 1 as shown

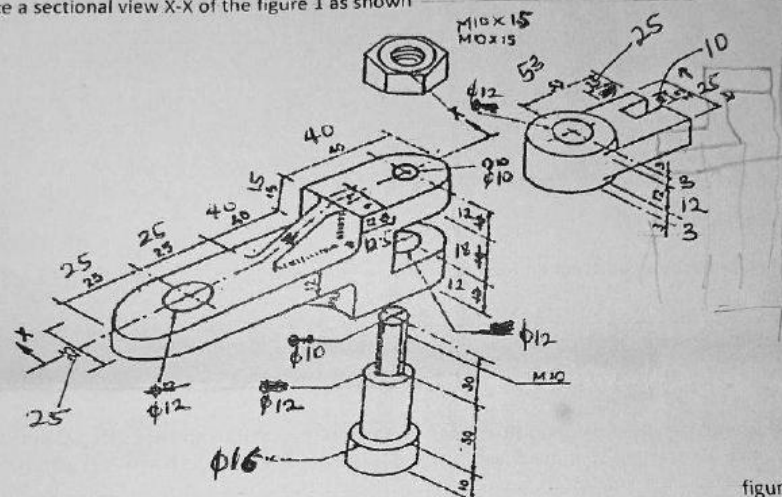


figure 1

QUESTION 2

15marks

Produce an orthographic projection of the figure 2 in first angle, taking the A- view as the front.

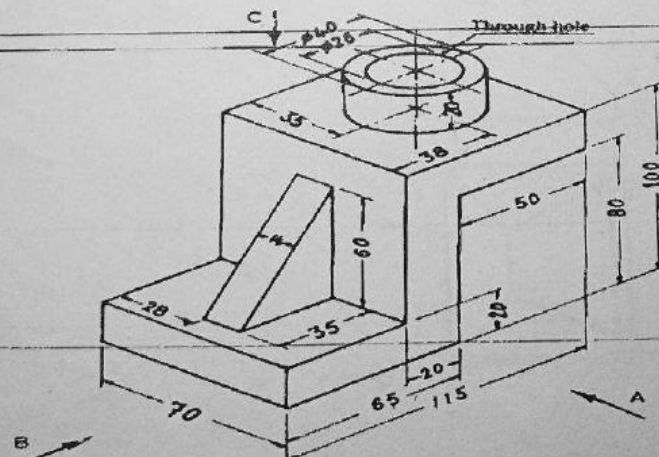
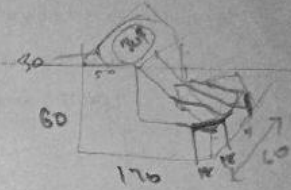
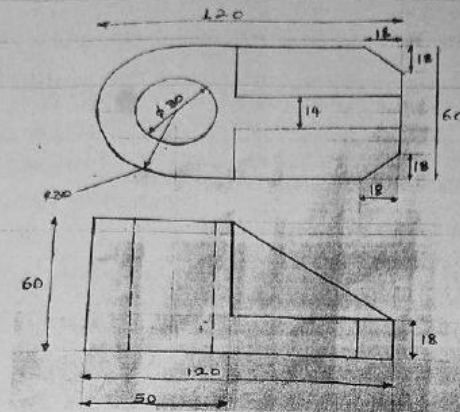


figure 2

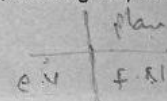
QUESTION 3 15marks

Produce an isometric equivalent of the third angle projection of a component shown in Figure 3.



QUESTION 4 15marks

- (a) In 3rd angle projection construct an M28 bolt with full thread length of 110mm, showing only the top and front views.
- (b) Construct its corresponding NUT as well, showing only the top and front views.



QUESTION 5 15marks

- (a) Figure 4 is a first angle projection of a truncated square pyramid, with base side 60 mm, height 70 mm and sloping face angle 45° starting 10 mm above base. Produce an auxiliary projection of the plan view.
- (b) From the auxiliary view, evaluate the true area of the truncated surface.

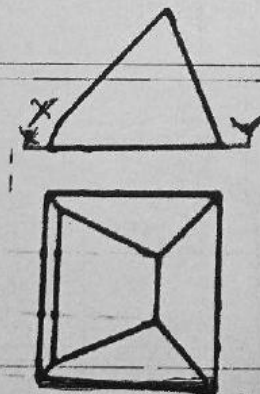
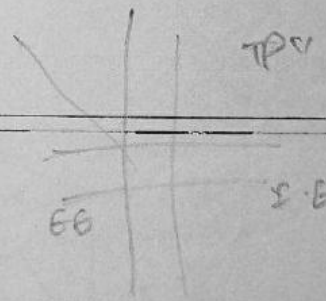


Figure 4.



Paper 4 of 6 — Mee 221

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DEPARTMENT OF MECHANICAL ENGINEERING
Second Semester Examination 2022/2023 Session
COURSE CODE AND TITLE: MEE 221 ENGINEERING DRAWING II

Instructions:

- (ii). Prepare of border and title block with free hand lettering carefully applied. – 10 marks

Answer Question One and any other two questions. That is a total of Three questions in all.
All dimensions are in mm if the unit is not indicated in the question.

QUESTION 1

30marks

- (a) Produce an assembly drawing of the four components as shown in figure 1.
- (b) Produce a sectional view X-X of the figure 1 as shown

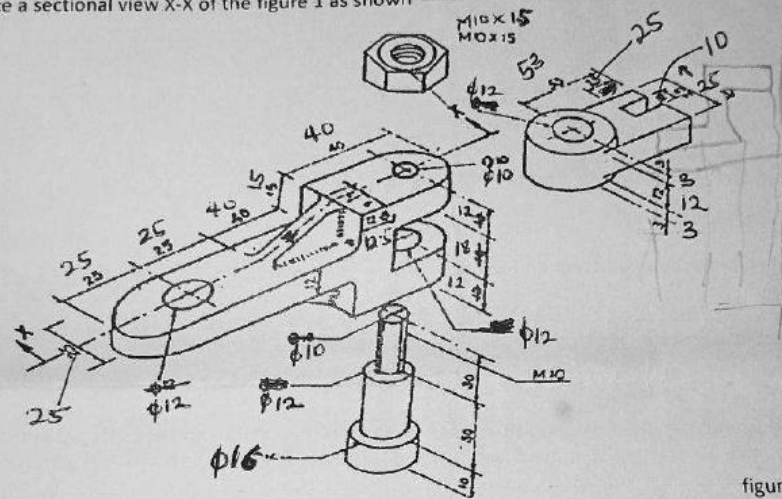


figure 1

QUESTION 2

15marks

Produce an orthographic projection of the figure 2 in first angle, taking the A- view as the front.

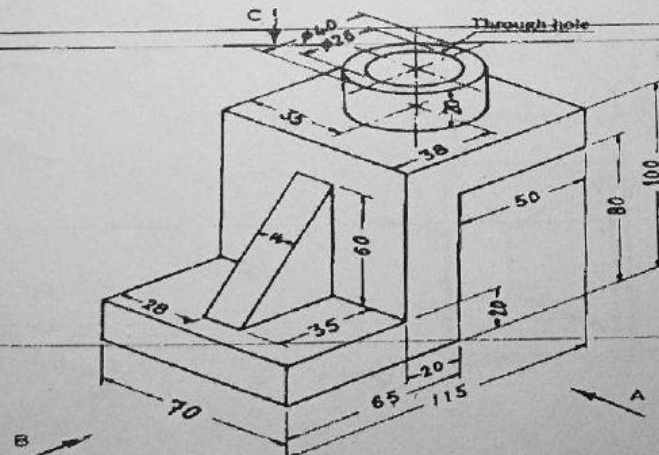
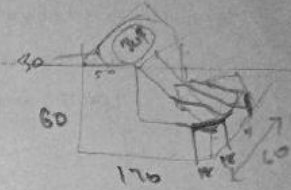
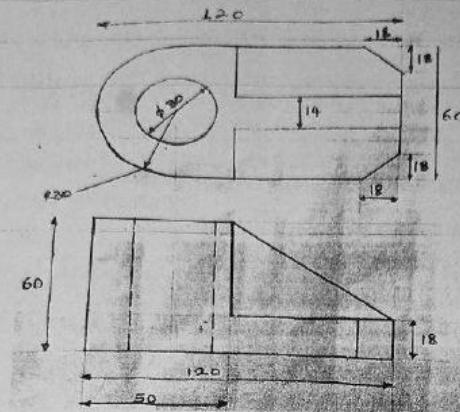


figure 2

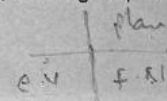
QUESTION 3 15marks

Produce an isometric equivalent of the third angle projection of a component shown in Figure 3.



QUESTION 4 15marks

- (a) In 3rd angle projection construct an M28 bolt with full thread length of 110mm, showing only the top and front views.
- (b) Construct its corresponding NUT as well, showing only the top and front views.



QUESTION 5 15marks

- (a). Figure 4 is a first angle projection of a truncated square pyramid, with base side 60 mm, height 70 mm and sloping face angle 45° starting 10 mm above base. Produce an auxiliary projection of the plan view.
- (b). From the auxiliary view, evaluate the true area of the truncated surface.

Thread length 110mm

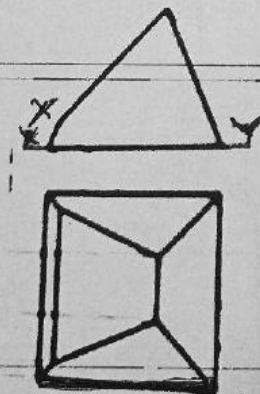
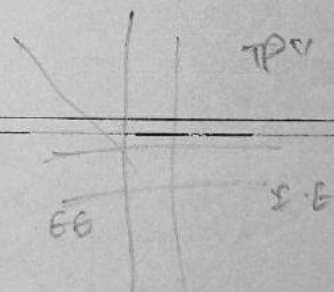


Figure 4.



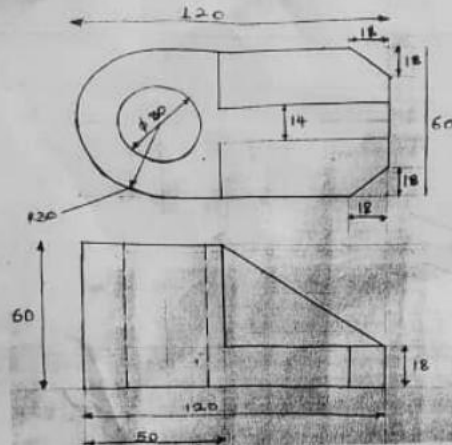
Paper 5 of 6 — MEE 221

Level 200 • Second Semester • PastQuestion

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QUESTION 3 15marks

Produce an isometric equivalent of the third angle projection of a component shown in Figure 3.



QUESTION 4 15marks

- In 3rd angle projection construct an M28 bolt with full thread length of 110mm, showing only the top and front views.
- Construct its corresponding NUT as well, showing only the top and front views.

QUESTION 5 15marks

- Figure 4 is a first angle projection of a truncated square pyramid, with base side 60 mm, height 70 mm and sloping face angle 45° starting 10 mm above base. Produce an auxiliary projection of the plan view.
- From the auxiliary view, evaluate the true area of the truncated surface.



Figure 4.



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DEPARTMENT OF MECHANICAL ENGINEERING
Second Semester Examination 2022/2023 Session
COURSE CODE AND TITLE: MEE 221 ENGINEERING DRAWING II

Instructions:

- (i). Prepare of border and title block with free hand lettering carefully applied. – 10 marks

Answer Question One and any other two questions. That is a total of Three questions in all.
All dimensions are in mm if the unit is not indicated in the question.

QUESTION 1 - 30marks

- (a) Produce an assembly drawing of the four components as shown in figure 1.
(b) Produce a sectional view X-X of the figure 1 as shown

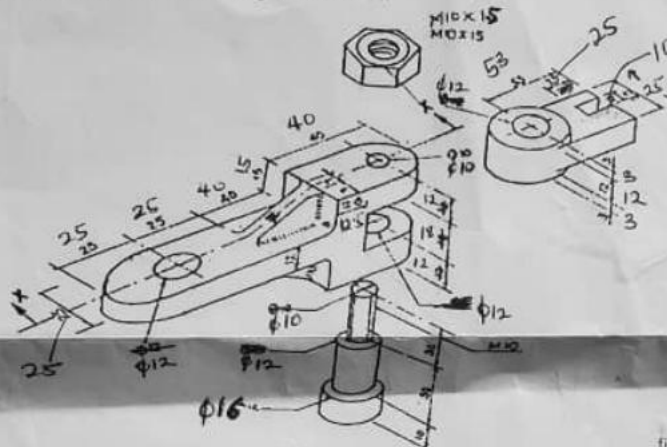


figure 1

QUESTION 2 15marks

Produce an orthographic projection of the figure 2 in first angle, taking the A- view as the front.

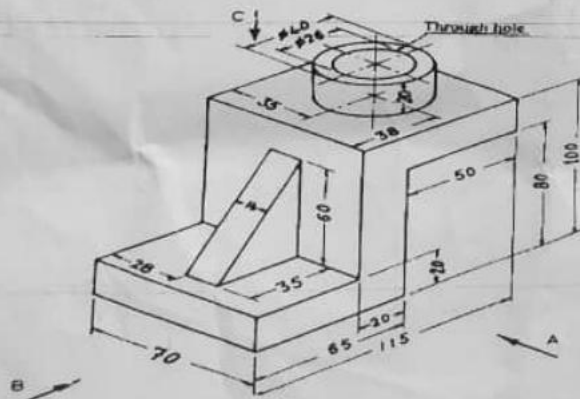


figure 2

Paper 6 of 6 — MEE 221 PAST QUESTION

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING

Second Semester Examination 2020/2021 Session

COURSE CODE AND TITLE: MEE 221

Engineering Drawing II

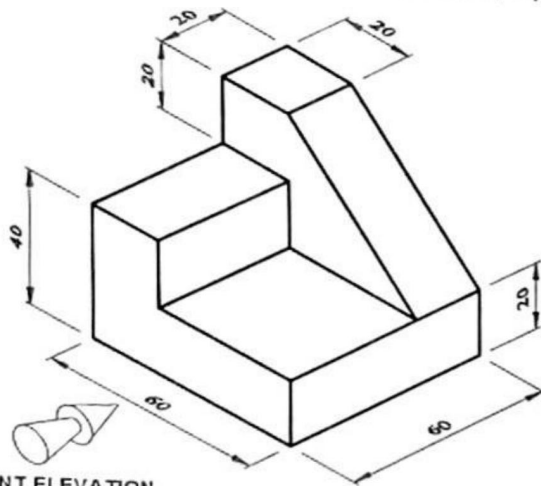
TIME: 3hrs

INSTRUCTION: Answer any three (3) Questions All Questions carry equal marks

Title block/Border line 5.5mks

QUESTION 1

- (a) Draw the isometric projection of the object as shown below. [10mks]
(b) Construct in 3rd angle projection the front, top and side views of the object below [11.5mks]



FRONT ELEVATION

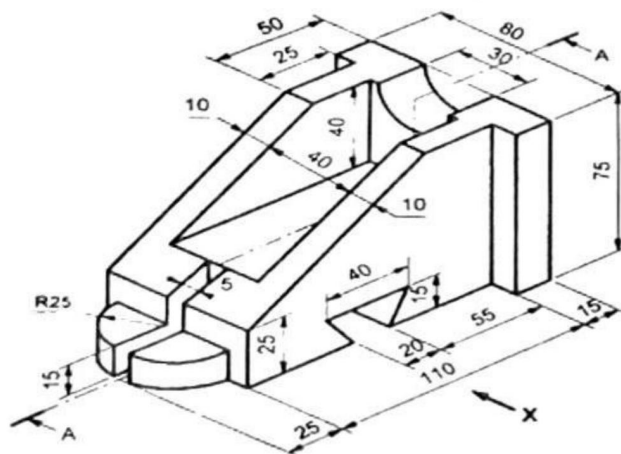
QUESTION 2

Draw the following views (i) the sectional front view along section A-A

[11.5mks]

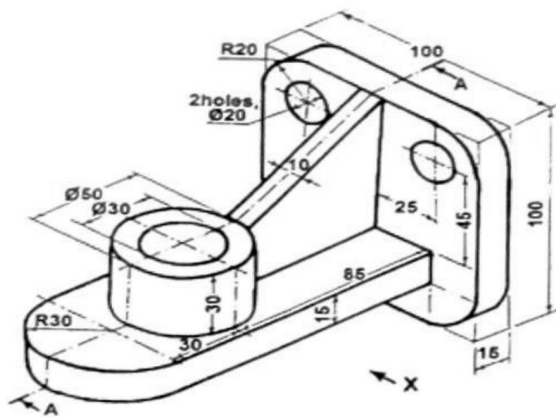
(ii) the top view only

[10mks]

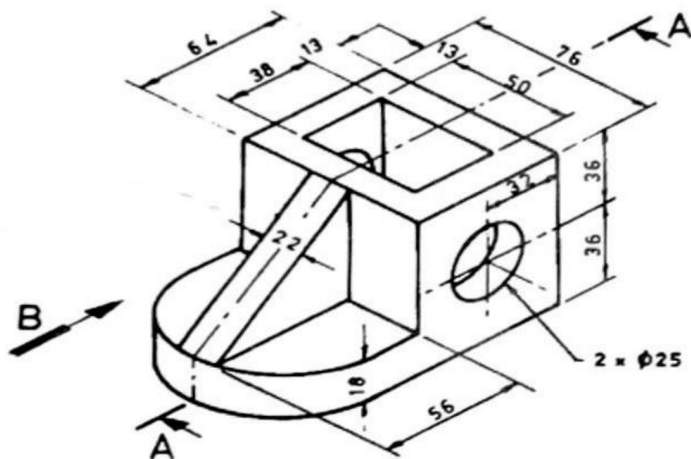


QUESTION 3

The figure below shows a bracket. Draw in 1st angle projection the front, top and side elevation viewing from point x [21.5mks]

**QUESTION 4**

- (i) Draw the sectional front view of the section A-A as show in the figure below [11.5mks]
 (ii) construct the front view only as indicated at point B [10mks]

**QUESTION 5**

- (i) Using a pencil and an eraser only, do a free hand sketching of the bearing block as shown below.
 (ii) Viewing from point x, freely construct the front, top and side views of the bearing block in 1st or 3rd angle projection [10mks and 11.5mks respectively]

